

DIPANJAN SAHA

+91 93301 45250 | dipanjansaha2003@gmail.com | linkedin.com/in/dipanjansahaa | github.com/dipanjansahaa

Summary

Associate Software Engineer at **Accenture**, working on **SAP ABAP** and **SAP BW/4HANA**. Strong in **data structures, backend development and SQL**, with experience building **ML-based and data-driven applications**. Skilled in **Python, C++, Java, REST APIs and cloud platforms (OCI)**, seeking roles focused on scalable systems and data-driven solutions.

Technical Skills

Languages: Python, C++, C, Java, SQL, JavaScript, HTML/CSS

SAP & Enterprise: SAP S/4HANA ABAP, Open SQL, Data Dictionary, ALV Reports, Smart/Adobe Forms, Enhancements, SAP BW/4HANA (Data Modeling, InfoProviders, ETL basics)

Backend & Databases: REST APIs, Flask, MySQL, Firebase

Cloud & AI: Oracle Cloud Infrastructure (OCI), Generative AI, LLMs, RAG, Vector Databases

Tools: Git, GitHub, Linux, VS Code, Eclipse, HANA Studio, Jupyter Notebook

Experience

Accenture

Jan 2026 – Present

Associate Software Engineer

Kolkata, India

- Prepared **SAP S/4HANA ABAP reports** (Classical and ALV) using Open SQL, internal tables, and joins to structure and retrieve enterprise data efficiently.
- Applied **performance optimization and modular programming practices** in ABAP, improving code readability and execution efficiency.
- Worked with **data dictionary objects, enhancements, and Smart/Adobe Forms**, gaining practical exposure to enterprise application development workflows.
- Contributing to **AESG PaTH Service Delivery project** for the State of New South Wales, involving **SAP ABAP development and SAP BW/4HANA data modeling and handling** for enterprise data workflows and reporting.

Projects

Call Simulation for D2D Communication | Python, RL, Simulation Modeling

May 2025 – Dec 2025

- Designed and simulated a **Device-to-Device (D2D) communication framework** to reduce base-station (BS) dependency and improve energy efficiency in cellular/IoT networks.
- Implemented a **Reverse Fair Scheduling (RFS)**-based channel allocation algorithm to optimize spectrum utilization and ensure interference-aware communication.
- Achieved **29.6% energy savings** compared to general BS communication while maintaining comparable throughput.

Emotion Echo: Multi-Modal Anxiety Detection System | Python, CNN, Firebase, IoT

Aug 2024 – Mar 2025

- Built a **multi-modal mental health monitoring system** integrating facial emotion recognition, heart rate sensing, and psychological inputs.
- Developed a **CNN model on FER2013 dataset**, achieving **82% accuracy** for real-time emotion classification.
- Designed a **weighted scoring algorithm** combining physiological and emotional signals, with Firebase integration for real-time data handling.

House Price Prediction Web App | Python, Flask, Scikit-learn, HTML/CSS/JS

Apr 2024 – Sep 2024

- Developed a **scalable machine learning-based web application** to predict housing prices using location, area, and room features.
- Built and deployed a **regression model** using Scikit-learn and integrated it with a Flask-based backend API.
- Designed a **user-friendly frontend interface** enabling real-time predictions for end users.

Education

B. P. Poddar Institute of Management & Technology

Aug 2021 – Jul 2025

Bachelor of Technology in Information Technology

CGPA: 8.14 / 10

Certifications & Publications

Certifications: OCI Generative AI Professional (2027), OCI Multicloud Architect (2027), OCI Data Science Professional (2027)

Publications: *Energy Efficiency Analysis of Device-to-Device Communication for the Sustainable Internet of Things under Diverse Network Scenarios* (2025); *Real-Time Anxiety Detection Using Emotion Echo: A Multi-Modal and Embedded Approach* (2024)